TTRPG Tools: Maps

This plugin was created to give you as much freedom as possible when designing RPG maps inside Obsidian.

Setup:

- **Install BRAT from the Community Plugins.**
- **Use BRAT to add a new community plugin, and use the URL: github.com/Jareika/zoom-map**

In a few weeks or months, the Obsidian team review might start, and then it could appear in the official Community Plugins list.

TTRPG Tools: Maps – Settings:

Storage:

- **Storage location by default:** This option lets you choose whether marker positions, view size, all layers (drawing, overlays, bases, etc.), drawings, map settings, and distance-measurement calibrations are stored in a JSON file or inline inside the note.

Layout:

- **Default width when wrapped:** If no width is set and wrap: true is enabled, the width defaults to 50%.

Interaction:

- **Mouse wheel zoom factor:** Controls how quickly you can zoom in and out using the mouse wheel.
- **Panning mouse button:** Choose whether you want to pan the map using the left or right mouse button
- **Hover popover size:** Defines the size of the popover that appears when hovering over a pin on the map. (Some themes override this with their own settings, such as the ITS theme.)
- **Force popovers without ctrl:** Lets you decide whether popovers should appear automatically or only when holding the Ctrl key.

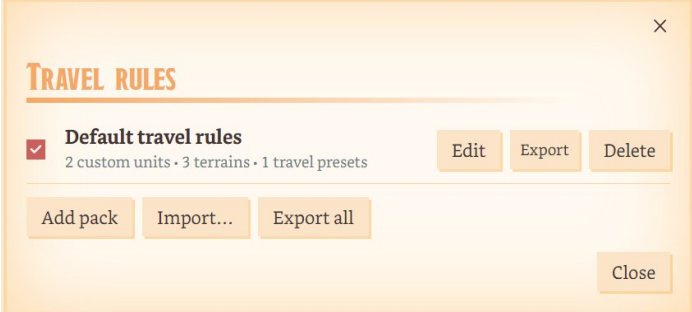
- **Open editor when placing pin from menu:** When enabled, the marker editor opens automatically whenever you place a new pin on the map. Pins can also be edited by right-clicking them.
- **Preferences:** Contains many advanced settings; explained later.

Ruler:

- **Line color and Line width:** The ruler is a distance-measurement tool. These settings control the color and thickness of the measurement line.

Travel rules:

- **Manage travel rules packs:** Opens a window where you can create and manage rule packs for different RPG systems. These packs define travel types, speeds, units, and terrain categories. Only active packs appear in the context menu.



Collections (base-bound):

- Here you can create, edit, and delete collections. A collection is a set of pin types that appears in the context menu only for the selected base map. This keeps the context menu organized even if you have hundreds of pins.

Marker icons (library):

- To prevent losing marker settings, you can export your entire marker icon library to a JSON file and load it again later. Image files themselves are **not** stored — they must still be located in the same place when reloading the library.

SVG icons:

- **Add SVG icon or sort the list:** Pick/add SVG icons or sort the list. Large SVG libraries may take a long time to render in the picker; adding icons is delayed until all previews are loaded.

- **SVG icon folder in vault:** Choose a folder and click “Ensure folder”.

- **Download icon packs:** There are two big free packs (Font Awesome and RPG Awesome). The plugin can download them, but you must unzip them manually (security reasons).
- **In the SVG icon list you can:**
  - enable a checkbox (so it's selectable in collections),
  - set a name (used in the context menu),
  - see a preview,
  - set color via text field + color picker,
  - set a default link
  - open SVG outline settings (color, line thickness, opacity),
  - set anchor size and rotation angle (anchor is the pivot/origin point).

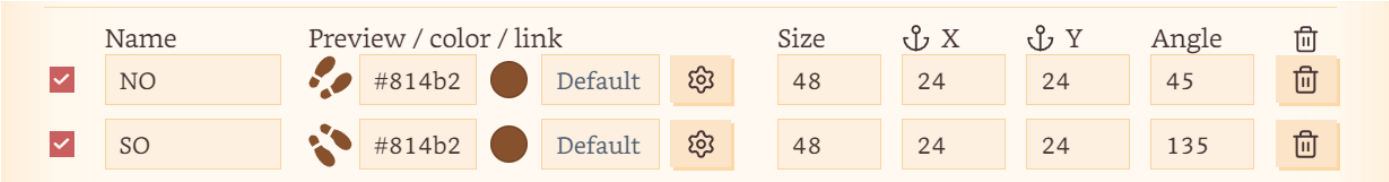
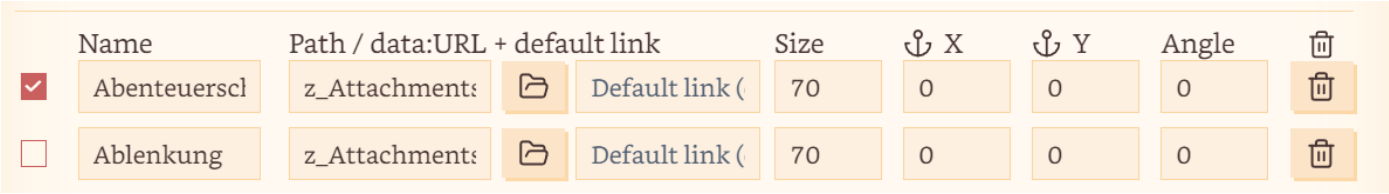


Image icons:

- **Add new icon or sort the list:** Add new icons or sort the list.
- In the list you can enable the checkbox, set a name, choose an image file, etc.



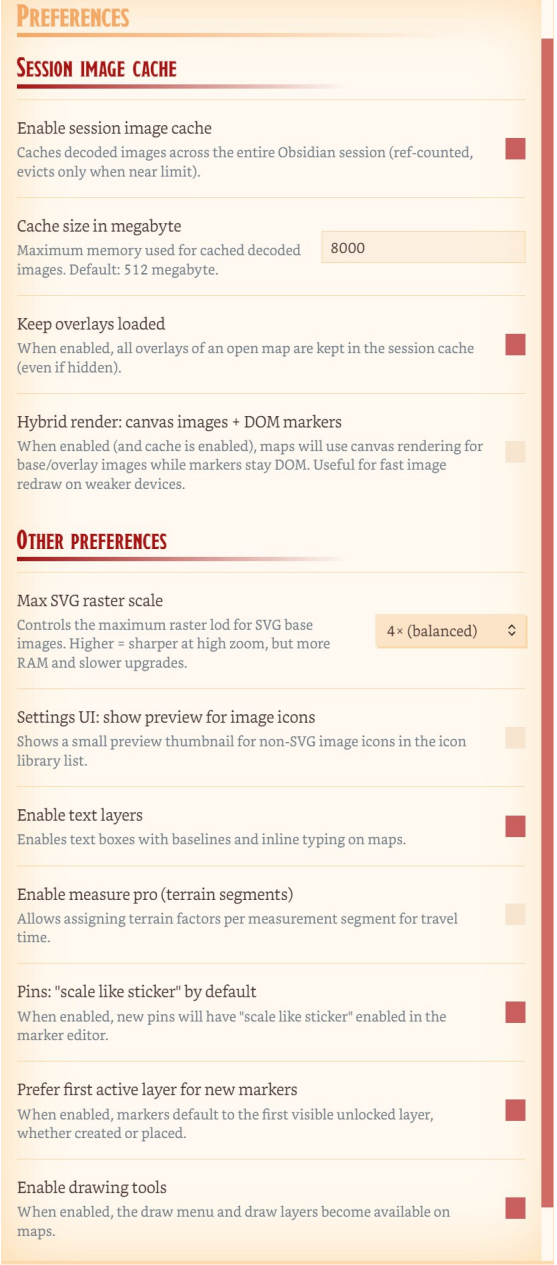
The Preferences:

Session Image Cache:

- **Enable session image cache:** Useful for weaker devices (e.g., tablets) with long load times. When enabled, images rendered via Canvas are kept in memory and load much faster. On Android tablets, Obsidian/Chromium/Electron may unload images when switching/closing tabs; this option prevents that.
- **Cache size in megabyte:** Example: a 4000×4000 image (~7.5 MB file) can use ~130 MB RAM. When the limit is reached, images from fully closed notes are removed first.
- **Keep overlays loaded:** Keep overlay images in memory even if

disabled, so re-enabling is faster.

- **Hybrid render:** Images in canvas, markers as DOM (performance compromise).



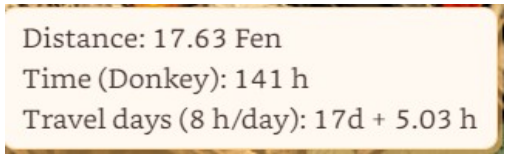
Other preferences:

- **Max SVG raster scale:** Select how high-res the map gets when zooming in. It can take a few seconds for higher resolution depending on system. I recommend converting SVG to WebP via context menu.
- **Settings UI: show preview for image icons:** Generates previews for image icons; with many icons this can impact performance.
- **Enable text layers:** Enables drawing text layers via context menu.
- **Enable measure pro (terrain segments):** Lets you pick a terrain type (from the active travel pack) per measurement segment to adjust displayed travel time based on terrain.
- **Pins: „scale like sticker“ by default:** If enabled, pins behave like stickers (their size scales with the map when zooming) instead of staying constant like classic pins.
- **Prefer first active layer for new markers:** New markers go into the first active marker layer.
- **Enable drawing tools:** Enables drawing tools in the context menu; you can draw shapes and store them in drawing layers.

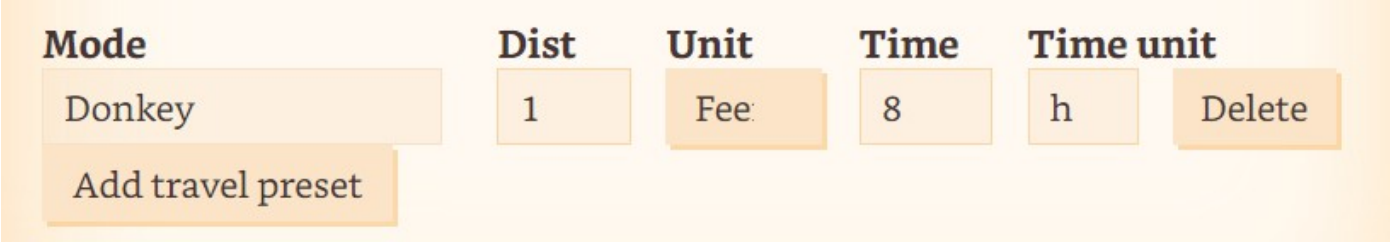
Travel rules:

Edit travel pack:

- **Name:** for the pack, e.g. “DnD 5e”.
- **Travel days:** Define maximum travel time per day.



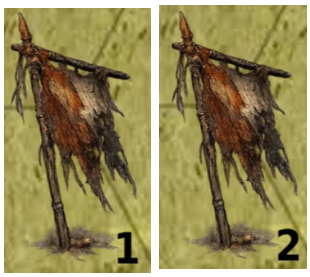
- **Custom units:** Define your own units instead of meters/kilometers/etc.
- **Terrains:** Define terrain types and their speed factor.
- **Travel time presets:** Define travel modes (example in handbook: donkey; 1 “Fen” takes 8 hours).



Collections:

Add oder Edit einer Kollektion:

- **Collection name**
- **Bindings (base images):** Decide in which maps this collection should be available.
- **Pins (from icon library):** Choose which pins appear under “Pins (base)” in the context menu.
- **Favorites (presets):** Add favorites from pins available in the collection.
- **Stickers:** Stickers are pin-like images with no links and are drawn under other pins; useful e.g. for a world map with a sticker pack of locations.
- **Swap pins:** Swap pins change image (and can change link) on right click. The handbook example: a banner icon that toggles between “1” and “2”. Useful if you revisit a location multiple times.



  - In the editor, you can set the name.
  - You can also choose whether it should be placed as a HUD pin by default or behave like a sticker.
  - A default link is only suitable if you always want the same link when placing that swap pin multiple times. To give swap pins different links per placed pin, edit the pin on the map with ALT + right click.
- **Party Pins:** Party pins help you find what is near the party, but they are also the most complex pins. Notes linked by pins should use **tags**

and/or **frontmatter properties** to use party pins effectively.

- Choose **name**, **icon**, and **marker layer**.
- Choose a **unit** via the travel pack.
- **Distances (comma separated):** Defines which party pin radii appear in the context menu.
- **Party note folder:** Where generated notes are stored.
- **Filter tags (optional, comma separated) and Filter properties (optional):** If a note in range has one of these tags/properties, it will appear in the party result with distance.
  - If that note has additional tags, the plugin searches the vault for other notes with those tags and writes a table (see example tables in the handbook).

Create a map inside a note:

The easiest way to create a map is the command: **TTRPG Tools: Maps: Insert new map…**

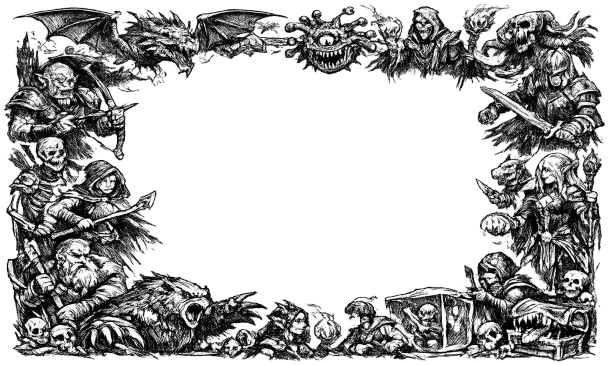
- The note must be in Edit mode; in Reading mode the command does not appear.
- You can assign a hotkey to the command.

The "Edit view" window (map setup):

- **Base images:** You can set base images here. Later you can switch between them on the map.
- **Overlays:** You can add overlays that will be shown above the base images (transparent images).
- **Marker JSON:** You can set the name/path of the markers JSON file (optional). If you don't set it, a markers JSON file will automatically be created with the same name as the first base image.
- **Marker layers (names):** Here you can create marker layers, so you

can later show/hide/lock markers.

- **View & layout:**
  - **Render mode:** Choose Canvas or DOM. For weaker systems and/or many overlays, Canvas is recommended.
  - **Min zoom/Max zoom:** How far you can zoom in/out.
  - **Wrap:** true or false: Allows text to flow around the map.
  - **Responsive:** true or false: Always sets width to 100% of the window and keeps the image aspect ratio.
  - **Width/Height:** Set the map viewport size. If the checkbox is enabled, the value is stored in YAML and the map will always reopen in that size (then “Resizable” isn’t permanently applied).
  - **Resizable:** If enabled, you can resize the map window by dragging in the note.
  - **Resize handle:** Only appears if Resizable is enabled. Options are Left, Right, Both, and Native. Native is on the right and provides the best look.
  - **Align:** Show the map left/right/center in the note.
  - **ID = optional:** Important for inline storage to uniquely identify the map data (also used for embeddings).
- **Viewport frame:** The viewport is the window that displays the map; you can decorate it with an image frame.
  - **Frame image (optional):** Select a frame image with transparency in the middle.
  - **Update viewport button:** Preview your frame/inset changes immediately without closing the window.



  - **Viewport insets unit:** Choose whether to enter insets as pixels or percent to fit the map into the frame.
  - **Inset Top/Right/Bottom/Left:** Shifts the map under the frame in that direction; use Update viewport to preview.
- **Save and Cancel**

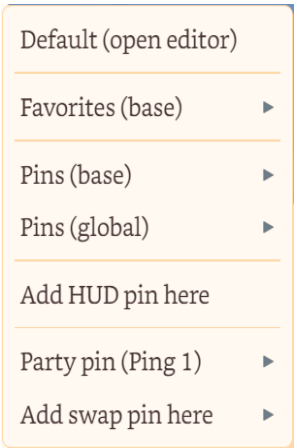
You can also open the Edit view window later from the context menu to

make changes.

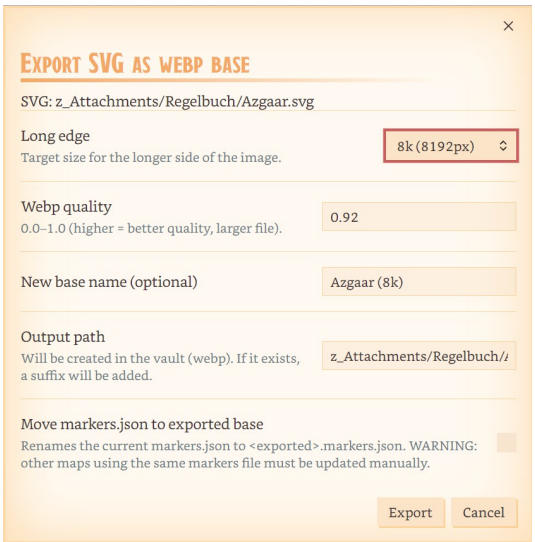
YAML note: You can write the YAML block yourself, but since it does not provide improvements or more options, the handbook does not document it.

The context menu (right click on the map):

- **Add marker here:** This menu entry lets you add different marker types (as long as you created them in Settings/Collections).
  - **Default (open editor):** Opens the marker editor and lets you create a pin.
  - **Favorites (base):** Add a favorite preset (defined in Collections).
  - **Pins (base):** Add a pin from the collection for the active base image.
  - **Pins (global):** If you created a global collection, you can place pins here.
  - **Add HUD pin here:** A HUD pin is independent from the map; it stays fixed in the viewport and does not get bigger/smaller when zooming.
  - **Party Pin:** Add a party pin (see Collections).
  - **Add swap pin here:** Place swap pins (see Collections).



- **Favorites (base):** Falls in den Kollektionen Favoriten definiert wurden erscheint dieser Punkt auch unter Add marker here.
- **Image layers:** Manage your image layers:
  - **Base:** Switch the base image.
  - **Overlays:** Toggle overlays on/off (multiple at once). For weaker systems, prefer Canvas.
  - **Export active SVG base as WebP:** If your base image is an SVG (e.g. from Azgaar Fantasy Map Generator), you

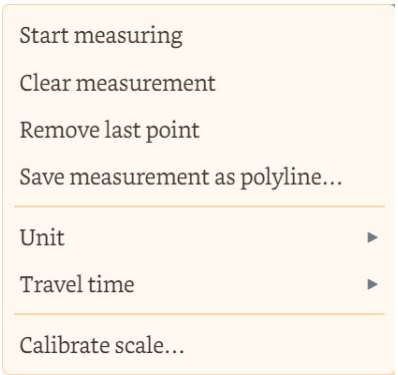


can convert it to WebP to avoid costly SVG re-rendering while zooming.

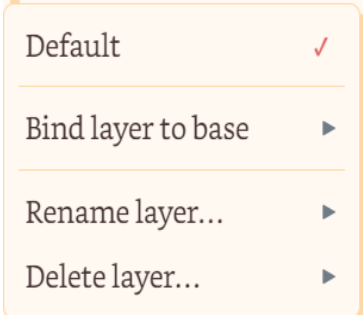
- **Long edge:** 4k / 8k / 12k. “Long edge” means the resolution is based on the longer side. Larger sizes take longer (12k can take 30 seconds or more depending on system).
- **WebP quality:** 1.0 is maximum quality.
- **New base name (optional):** Choose a new name.
- **Output path:** Set a new path or keep the current one.
- **Move markers.json to export base:** Renames/moves the old markers JSON from the SVG base to the new WebP base markers JSON so names match.

- **Draw layers:** This is where you can add, activate, rename, and delete drawing layers.
- **Delete base... und Delete overlay...:** This allows you to remove base images and overlay images from the map.
- **Add layer:** Creates a new base image or overlay image.

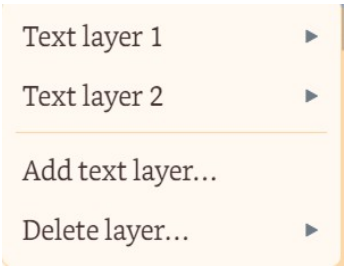
- **Measure:** You can measure distances and calculate travel times.
  - **Start measuring:** Start a new measurement (you must calibrate at least once before).
  - **Clear measurement:** Clear the measurement.
  - **Remove last point:** Deletes the last waypoint.
  - **Save measurement as polyline:** Saves the path as a line so you can style it afterwards.
  - **Unit:** Choose the unit (various options).
  - **Travel time:** Choose your travel method from enabled Travel Packs
  - **Calibrate scale...:** Click two points on the map and enter how far apart they are.



- **Marker layers:** Marker layers let you show/hide/lock markers:
  - Clicking a layer cycles through states: **Active** → **Hidden** → **Locked**.
  - **Bind layer to base:** Bind a marker layer to a base image; when you switch bases, the layer becomes hidden.
  - **Rename layer.../Delete layer...**

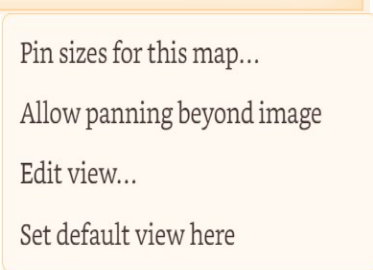


- **Text layers:** Text layers are boxes on the map where you can draw baselines and write on them (useful with swap pins for simple character sheets).
  - **Add text layer...:** Draw a text layer box (click+hold left mouse, drag a rectangle).
    - **Name:** The layer name.
    - **Allow angled baselines:** If enabled, you can draw baselines at an angle.
    - **Font:** Set font family, size, color, weight, italic, letter spacing.
    - **Layout:**
      - **Line height (px):** Set line height; otherwise it is auto-calculated from font size.
      - **Padding left/right (px):** Where text starts/ends on the baseline.
      - **Apply style to all text layers:** If you already have multiple text layers, you can apply this layer's style to all of them.
    - **Save and Cancel**
  - **Delete layer...:** Deletes text layers.



- **Draw:** Lets you draw shapes (rectangle, circle, line, polygon) and style them. The "Edit Drawing" menu opens after drawing.
  - **Label** can be set.
  - **Stroke:** solid/dashed/dotted; color, thickness, opacity.
  - **Fill:** solid/striped/cross/wavy; color, opacity, spacing, angle, line thickness, stripe opacity.
  - **Save/Delete/Cancel**

- **Options:** Gives various options for the map.
  - **Pin sizes for this map...:** Adjust pin sizes on this map independently of global icon settings. Also available when right-clicking a pin.
  - **Allow panning beyond image:** Normally you can pan only until the map edge, but



sometimes you need more.

- **Edit view...:** Re-open the Edit view window to change settings or add a frame.
- **Set default view here:** If enabled, the map always opens at this position and zoom. Currently this can only be removed via YAML.

Glossar:

Base	The active base image (e.g., day/night version).
Overlay	An additional image above the base (often transparent; e.g. labels/roads).
Marker / Pin	Clickable symbol on the map, optionally with link and tooltip.
Marker Layer	A layer that groups markers; can be visible/hidden/locked.
Bind layer to base	Ties a marker layer to a base image so it's only visible there.
HUD Pin	Pin positioned relative to the viewport; does not scale with zoom.
Sticker	Decoration without link, below normal pins; good for location marker sets.
Swap Pin	Pin with multiple states/icons; right-click cycles states (and optionally link).
Party Pin	Special pin that evaluates markers/notes in a radius and generates/updates a party note.
Ruler / Measure	Measure mode: click points to measure distance and show it live.
Calibrate scale	Calibrates the scale (e.g. m/px) so measurements become real distances.
Custom units	Your own units instead of m/km (e.g. "Hex").
Terrains	Terrain types with factor to adjust travel time per segment.
Travel time preset	Distance → time preset (e.g. 1 mi = 4 h).
Session image cache	Keeps rendered images in RAM so switching/reloading is faster.
Render mode (DOM/Canvas)	DOM = classic image display; Canvas can be smoother on weaker devices.
Viewport frame	Frame image above the map; insets define the visible map area
Responsive	Map adapts to width and keeps aspect ratio; interactions are limited.
Edit view	Setup window for bases/overlays/marker layers and layout.
YAML (zoommap block)	Configuration inside the code block; corresponds to what the View Editor generates.
markers.json	Storage file containing markers, layers, overlays/bases, scale, etc. (The path can be set in the View Editor; otherwise it is derived automatically from the first base image.)
Storage (json / note)	Where the map data is stored: either as a JSON file or inline inside the note.
ID (optional)	A unique identifier, especially important for inline storage (and embeddings) to ensure multiple maps can be matched correctly.
Wrap	Allows text to flow around the map (e.g., when aligned left/right).
Align	Align the map left/center/right inside the note.
Resizable	The map can be resized directly inside the note (drag/resize).
Resize handle	Type of resize handles (Left / Right / Both / Native).
Icon library	Global library of all pin icons (SVG or image), used to populate "Pins (base/global)".
Collections (base-bound)	Collections that define which Pins / Favorites / Stickers / Swap Pins / Party Pins appear in the context menu — depending on the active base image.
Global collection	A collection without base bindings; its items are available everywhere (shown as "Pins

	(global)”).
Favorites (presets)	Predefined marker templates you can place quickly
Sticker preset	A predefined sticker (image + size) made available through collections.
Swap frame	A “state” of a Swap Pin (icon + optional link). Right-click cycles through the states.
ALT-Right-click (Swap Pin)	Opens the editor / advanced settings (e.g., to assign individual links to specific Swap Pins).
Party note	The generated note created and updated by a Party Pin.
Party radius / range	The distance within which markers/notes are considered “in range”.
Filter tags / Filter properties	Conditions that determine which notes appear in the Party results (tags / frontmatter).
Travel pack	A package of rules: custom units + terrains + travel presets. Only enabled packs appear in the context menu.
Travel days	Option to display travel time additionally as “days + remaining time” (based on “max travel time per day” from the pack).
Measure pro (terrain segments)	Choose a terrain factor for each segment between points to adjust travel time per segment.
Draw laye	Layer for drawings (can be activated, renamed, or deleted).
Polyline / Polygon / Rect / Circle	Drawing types (line, polygon, rectangle, circle)
Pattern fill	Pattern fill for shapes (e.g., striped / cross / wavy).
Text layer (Box)	A rectangular area on the map where text can be written along defined lines.
Baseline	The text line (start/end point) on which the text sits.
Allow angled baselines	Baselines can be drawn at an angle (not only horizontally).
Line height / Padding left/right	Layout parameters for text input lines in the text layer (line height / inner spacing).
Session image cache	Keeps rendered/decoded images in RAM to speed up switching/reloading (useful on tablets/Android).
Keep overlays loaded	Overlays remain in memory even when hidden (faster re-display)
Hybrid render (Canvas images + DOM markers)	Images are rendered on the canvas, markers as DOM elements (performance compromise).
Export active SVG base as WebP	Converts the active SVG base to a high-resolution WebP to avoid re-rendering the SVG during zooming.